Au+Au 3.85 GeV: Λ–^{4}He CF in four m_T slices

Centrality 0–10%, KFParticle

CF: 20 MeV/c; display 0-0.3 GeV/c; normalization 0.30-0.40 GeV/c.

The boundaries are selected at stored 0.05 GeV/c^2 edges, near SE signal quartiles in $0 \leq k^* < 0.3$ GeV/c. The correlation shapes are not used to optimize the boundaries. The same slices are used for both methods, SE, ME and sidebands.

Slice	m_T [GeV/c^2]	SE (0–0.3)	ME (0–0.3)	SE norm	ME norm
1	2.40–2.60	12175	436803	20366	759268
2	2.60–2.70	17473	625983	23059	843567
3	2.70–2.80	14273	515692	16562	611419
4	2.80–3.90	14993	541619	16288	590353

Method 5 is an approximation, not an m_T-resolved mass fit. The available histograms have no joint (k^*, m_T, M_Λ) distribution. The same m_T-integrated, centrality-0–10% $P_{SE}(k^*)$ and $P_{ME}(k^*)$ are applied to all four slices. Purity grid: 50 MeV/c; five-bin sliding window: 250 MeV/c in the interior. Each original 5 MeV/c count is weighted before summing to 20 MeV/c.

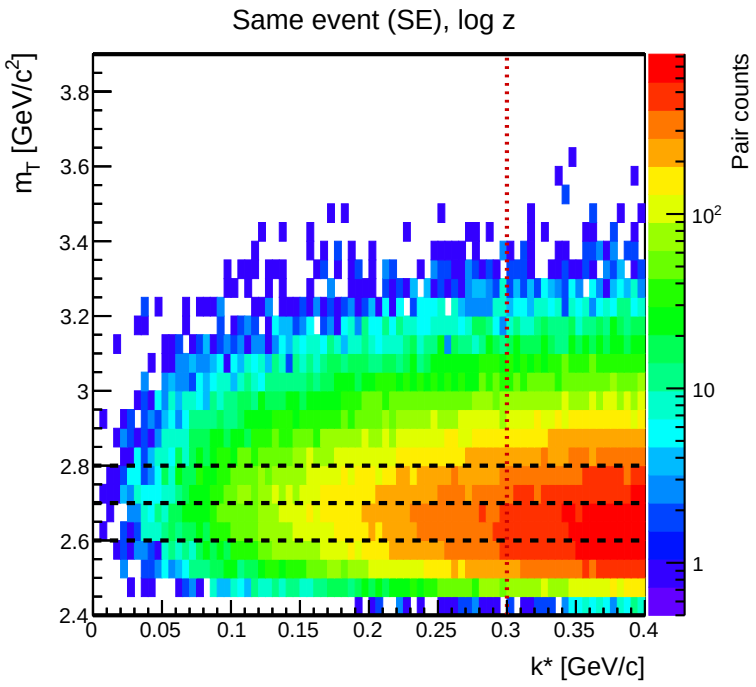
Constant purity: $P = 0.96721065$ (the same inclusive proxy as before), with $C_{corr} = [C_{sig} - (1-P)C_{SB}]/P$. Sidebands and normalization are evaluated separately in each m_T slice.

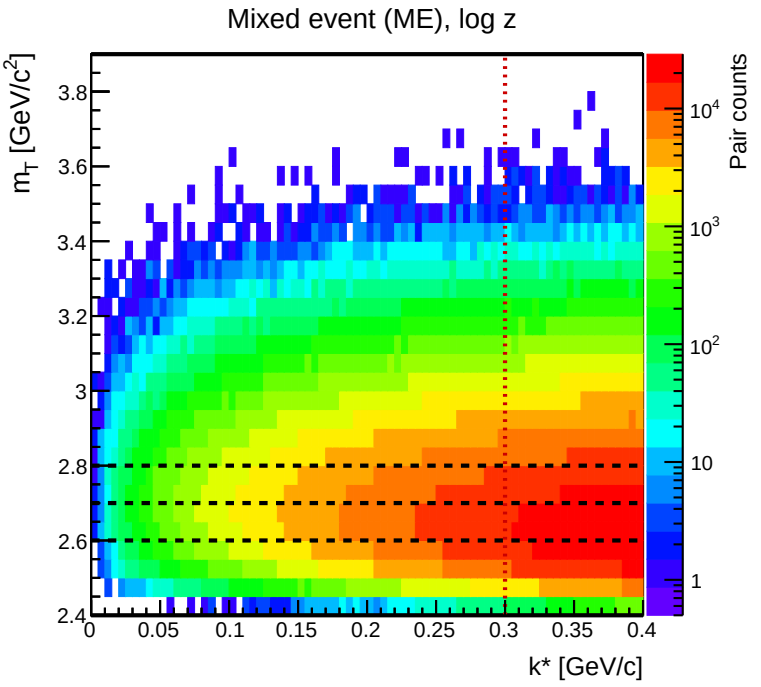
Approximate errors include shared purity reuse and normalization. Covariances between mass-fit backgrounds and observed counts, distinct overlapping fits, m_T counts and inclusive purity, and reused event/mixed pairs are not available. Constant-P errors are conditional on fixed P. Sparse/zero-SE bins do not have Poisson confidence intervals. The smallest SE signal bins contain only a few pairs despite the large slice totals. Exact candidate mass gates and whole mass-bin fit integrals differ slightly.

2021 P24iy only; 26,956/26,994 subjobs in this partial merge. The input ROOT and all production cuts are unchanged. Full provenance, statistics and caveats are in the adjacent README and JSON files.

Λ-^{4}He: m_T split map | centrality 0-10%

Signal: ±3σ; original bins; horizontal lines: slice boundaries; normalization 0.30-0.40 GeV/c





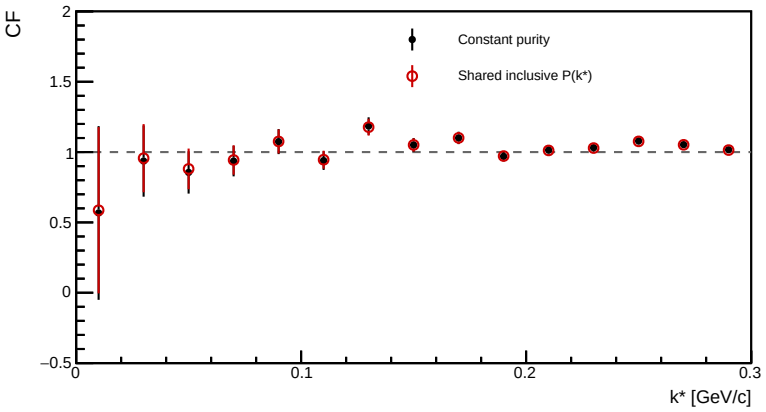
Black horizontal lines: supplied m_T edges. Red vertical line: normalization start. No cut or reweighting is applied by this renderer.

Λ-^{4}He: m_T-selected correlation functions

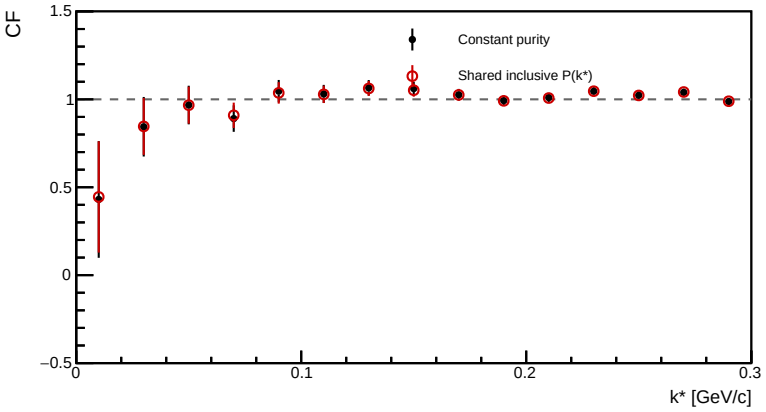
Signal: ±3σ; normalization 0.30-0.40 GeV/c; no hidden clipping of finite errors

Shared inclusive P(k*) (Method 5 approximation): purity is NOT fitted separately in each m_T slice

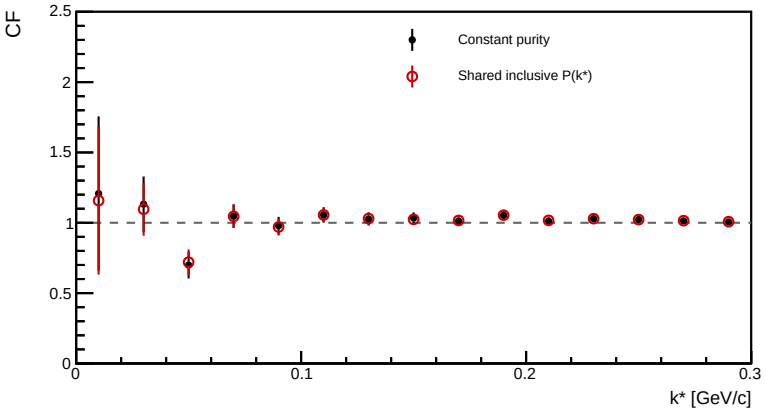
m_T = 2.40-2.60 GeV/c^2



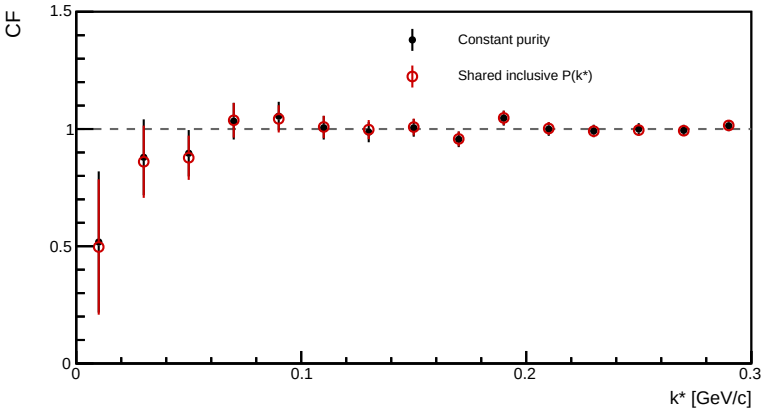
m_T = 2.60-2.70 GeV/c^2



m_T = 2.70-2.80 GeV/c^2



m_T = 2.80-3.90 GeV/c^2



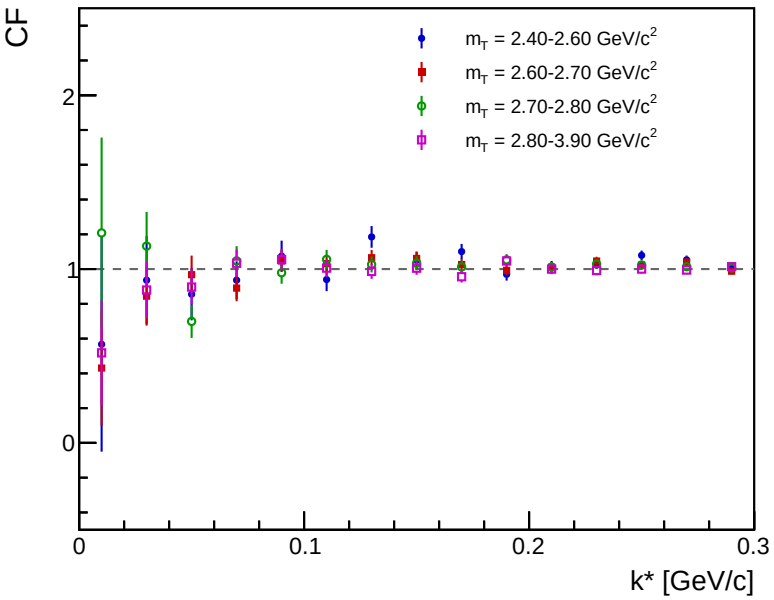
Invalid / missing bins are omitted, not set to zero. Normalization is shaded where visible.

Λ-^{4}He: m_T-selected correlation functions

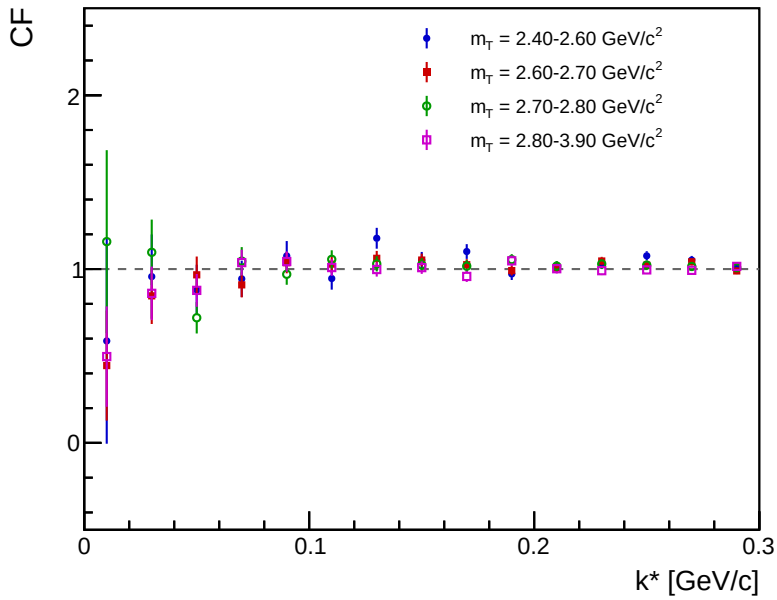
Signal: ±3σ; normalization 0.30-0.40 GeV/c; no hidden clipping of finite errors

Shared inclusive P(k*) (Method 5 approximation): purity is NOT fitted separately in each m_T slice

Constant purity



Shared inclusive P(k*)
(Method 5 approximation)



The same supplied CF values are reused; this renderer performs no fit or background correction.

Au+Au 3.85 GeV: Λ–^{4}He CF in four m_T slices

Centrality 0–10%, KFParticle

CF: 20 MeV/c; display 0-0.3 GeV/c; normalization 0.30-0.40 GeV/c.

The fixed boundaries are specified in YAML and coincide with stored 0.05 GeV/c^2 edges. They are not moved to equalize counts. Only the configured m_T range is included; tails outside it are excluded. Excluded signal pairs for $0 \leq k^* < 0.3$ GeV/c: below range SE=0, ME=0; above range SE=237, ME=7842.

The same slices are used for both methods, SE, ME and sidebands.

Slice	m_T [GeV/c^2]	SE (0–0.3)	ME (0–0.3)	SE norm	ME norm
1	2.40–2.60	12175	436803	20366	759268
2	2.60–2.80	31746	1141675	39621	1454986
3	2.80–3.00	12532	453695	13637	495936
4	3.00–3.20	2224	80082	2392	85425

Method 5 is an approximation, not an m_T-resolved mass fit. The available histograms have no joint (k^*, m_T, M_Λ) distribution. The same m_T-integrated, centrality-0–10% $P_{SE}(k^*)$ and $P_{ME}(k^*)$ are applied to all four slices. Purity grid: 50 MeV/c; five-bin sliding window: 250 MeV/c in the interior. Each original 5 MeV/c count is weighted before summing to 20 MeV/c.

Constant purity: $P = 0.96721065$ (the same inclusive proxy as before), with $C_{corr} = [C_{sig} - (1-P)C_{SB}]/P$. Sidebands and normalization are evaluated separately in each m_T slice.

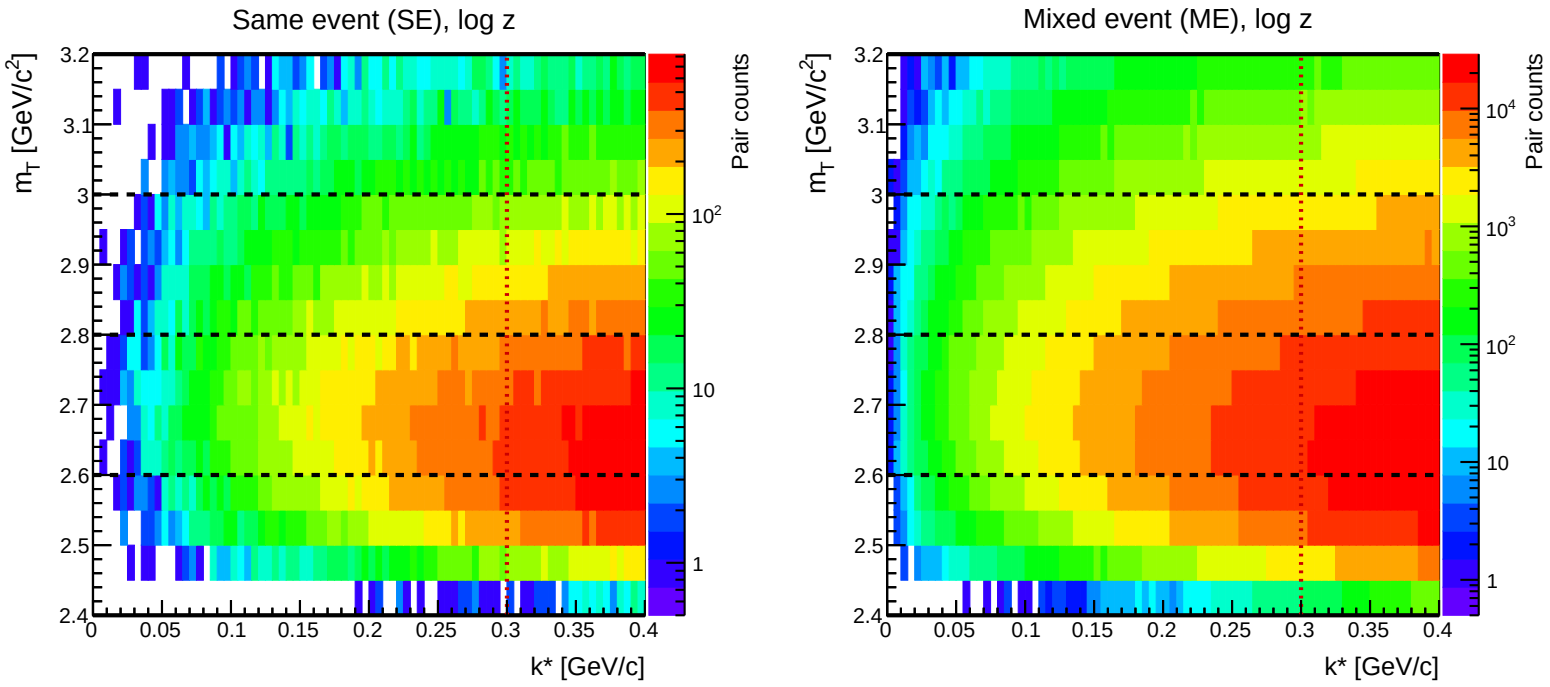
Approximate errors include shared purity reuse and normalization. Covariances between mass-fit backgrounds and observed counts, distinct overlapping fits, m_T counts and inclusive purity, and reused event/mixed pairs are not available. Constant-P errors are conditional on fixed P. Sparse/zero-SE bins do not have Poisson confidence intervals. The smallest SE signal bins contain only a few pairs despite the large slice totals. Exact candidate mass gates and whole mass-bin fit integrals differ slightly.

2021 P24iy only; 26,956/26,994 subjobs in this partial merge. The input ROOT and all production cuts are unchanged. Full provenance, statistics and caveats are in the adjacent README and JSON files. **Masked points (not zero):**

mT 3.00-3.20, constant_purity, k* 0.00-0.02. Zero ME denominators / invalid normalization are not filled.

Λ-^{4}He: m_T split map | centrality 0-10%

Signal: ±3σ; original bins; horizontal lines: slice boundaries; normalization 0.30-0.40 GeV/c



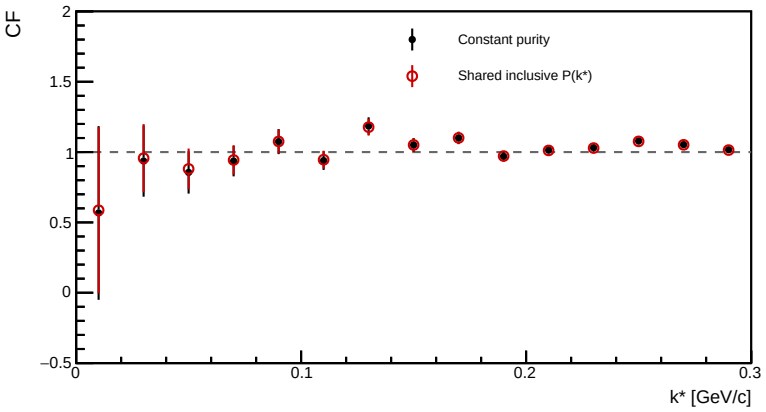
Black horizontal lines: supplied m_T edges. Red vertical line: normalization start. No cut or reweighting is applied by this renderer.

Λ-^{4}He: m_T-selected correlation functions

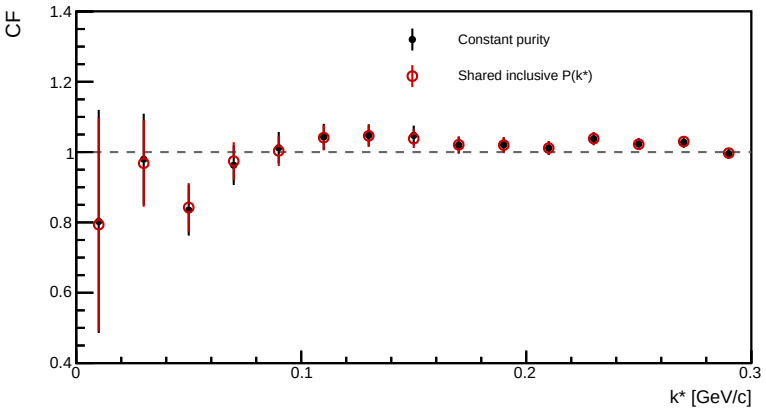
Signal: ±3σ; normalization 0.30-0.40 GeV/c; no hidden clipping of finite errors

Shared inclusive P(k*) (Method 5 approximation): purity is NOT fitted separately in each m_T slice

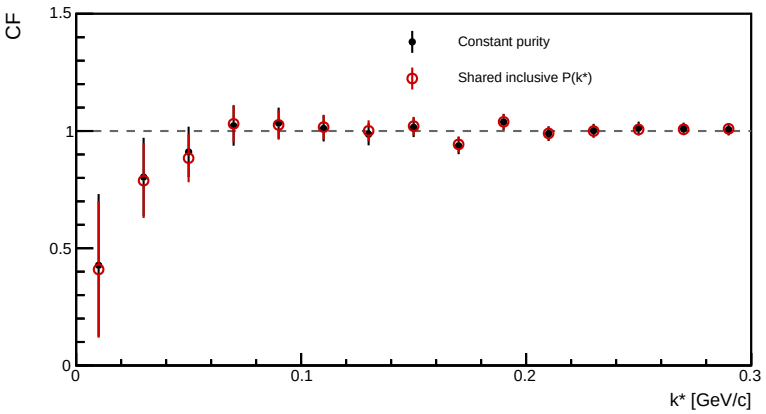
m_T = 2.40-2.60 GeV/c^2



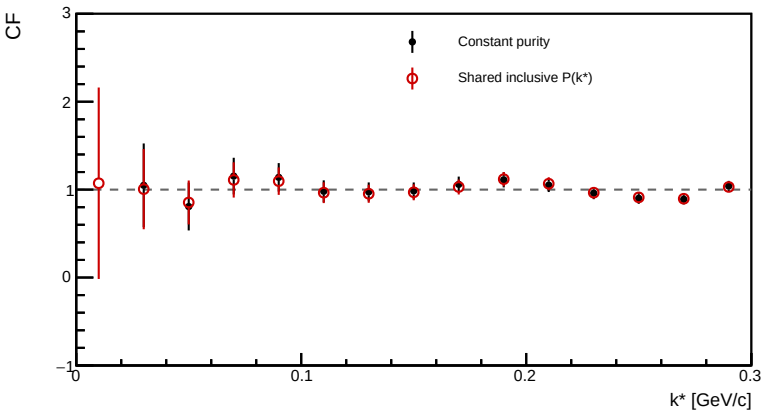
m_T = 2.60-2.80 GeV/c^2



m_T = 2.80-3.00 GeV/c^2



m_T = 3.00-3.20 GeV/c^2



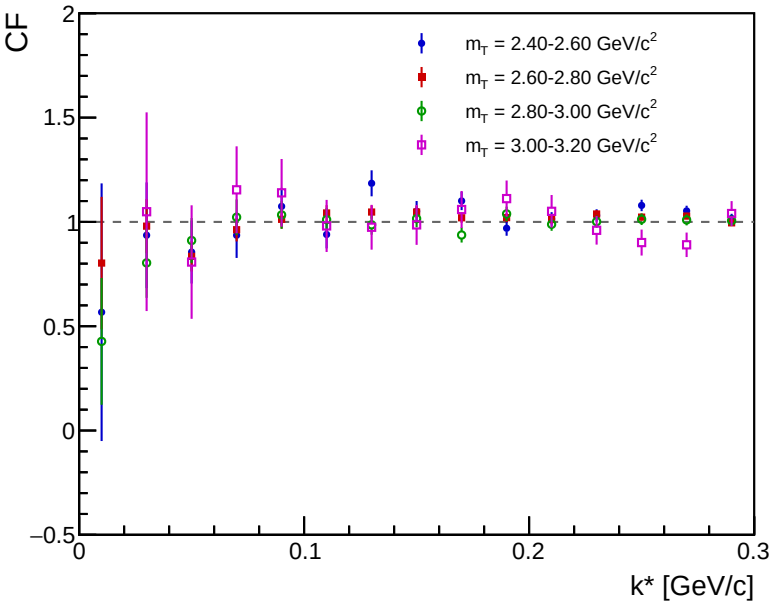
Invalid / missing bins are omitted, not set to zero. Normalization is shaded where visible.

Λ-^{4}He: m_T-selected correlation functions

Signal: ±3σ; normalization 0.30-0.40 GeV/c; no hidden clipping of finite errors

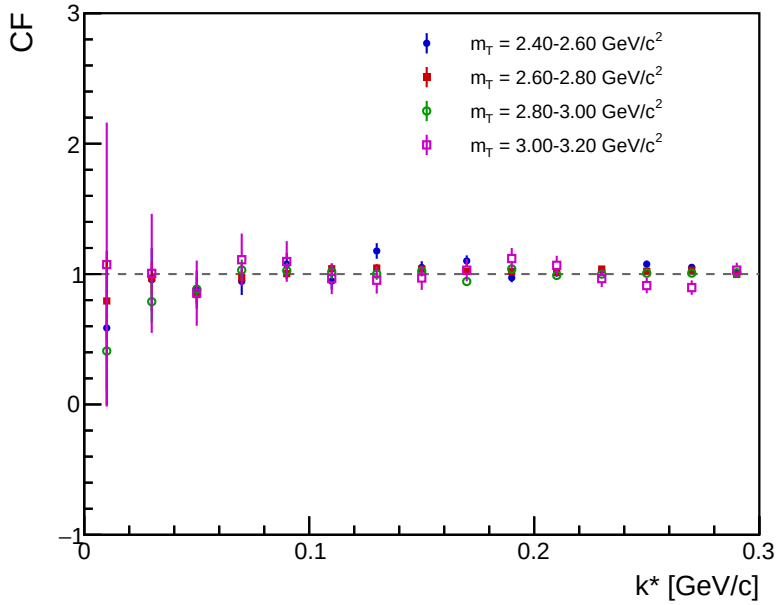
Shared inclusive P(k*) (Method 5 approximation): purity is NOT fitted separately in each m_T slice

Constant purity



Shared inclusive P(k*)

(Method 5 approximation)



The same supplied CF values are reused; this renderer performs no fit or background correction.